AI4Chemical Sciences Bootcamp — Introduction

An introduction to AI for the Chemical Sciences — Bootcamp overview

J. Schleinitz · Monday, August 10, 2026 · RSC 275, Caltech

Morning Agenda

I. **AI & ML today**
II. Bootcamp Overview
III. Bootcamp Objectives & Roadmap
IV. Logistics

Artificial Intelligence – Definition

AI is the capability of computational systems to perform tasks typically associated with human intelligence, such as:

- Learning
- Reasoning
- Planning
- Perception
- Communication
- …

Artificial Intelligence Forecasts?

When will there be a 50% chance that Human-level Artificial Intelligence exists?

Our World in Data

Timelines of 812 AI experts surveyed in three studies between 2018 and 2022.

1) Timelines of **356 AI experts**, surveyed **in 2022** by **Katja Grace et al.:**

The experts were asked when unaided machines will be able to accomplish every task better and more cheaply than human workers.

2) Timelines of **296 AI experts**, surveyed **in 2019** by **Baobao Zhang et al.:**

The experts were asked when machines will collectively be able to perform more than 90% of all tasks that are economically relevant better than the median human paid to do that task.

3) Timelines of **165 AI experts**, surveyed **in 2018** by **Gruetzemacher et al.:**

The experts were asked when AI systems will collectively be able to accomplish 99% of tasks that humans are paid to do at or above the level of a typical human.

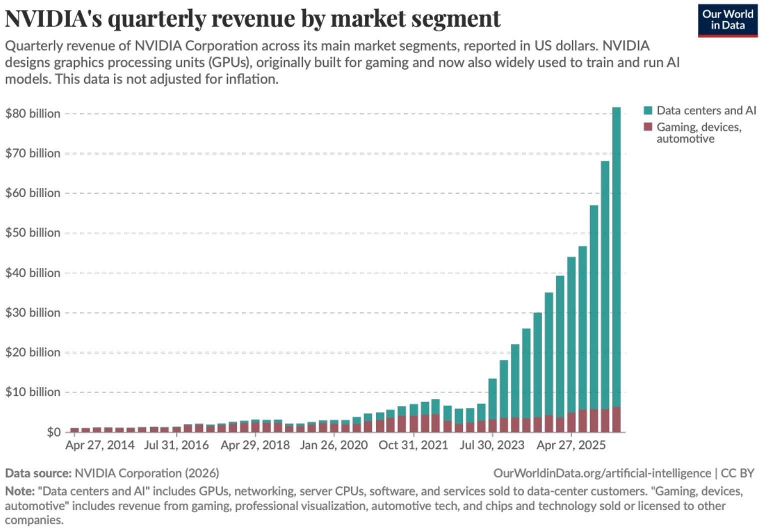
Full details on all studies and the questions that the AI experts were asked can be found in the text at OurWorldInData.org/AI-timelines.

OurWorldinData.org – Research and data to make progress against the world's largest problems.

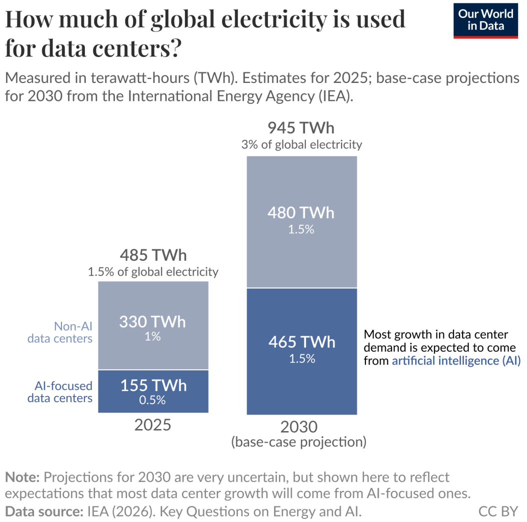
https://ourworldindata.org/ai-timelines

Artificial Intelligence and Society

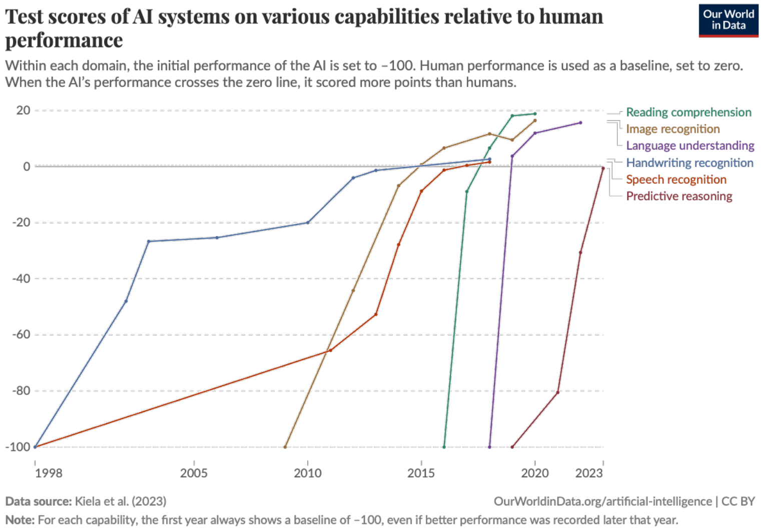
Economic growth



Energy Consumption

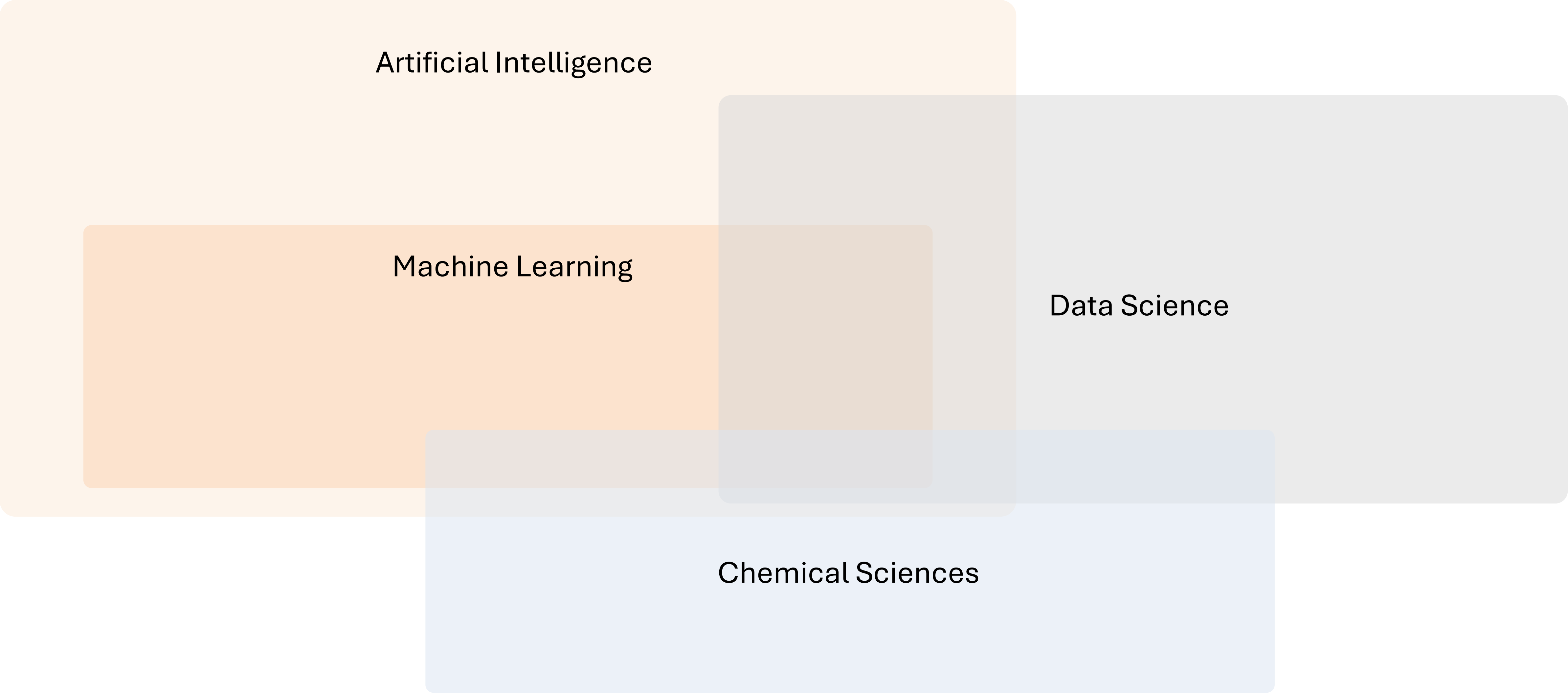


Benchmark against human performance



https://ourworldindata.org/artificial-intelligence
https://ourworldindata.org/how-much-energy-do-data-centers-and-artificial-intelligence-use

Artificial Intelligence & Machine learning



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Bootcamp Team

J. SCHLEINITZ
Bootcamp lead & organizer

S. WAHLER
Supervised Learning · LLM Assistants

M. MURPHY
MLIPs & Graph Neural Networks

C. LIU
Diffusion & Generative Models

M. YUSOV
Model Interpretability

A. SCHMIDT
Design of Experiments · Data Stewardship

B. REINER
Agentic AI

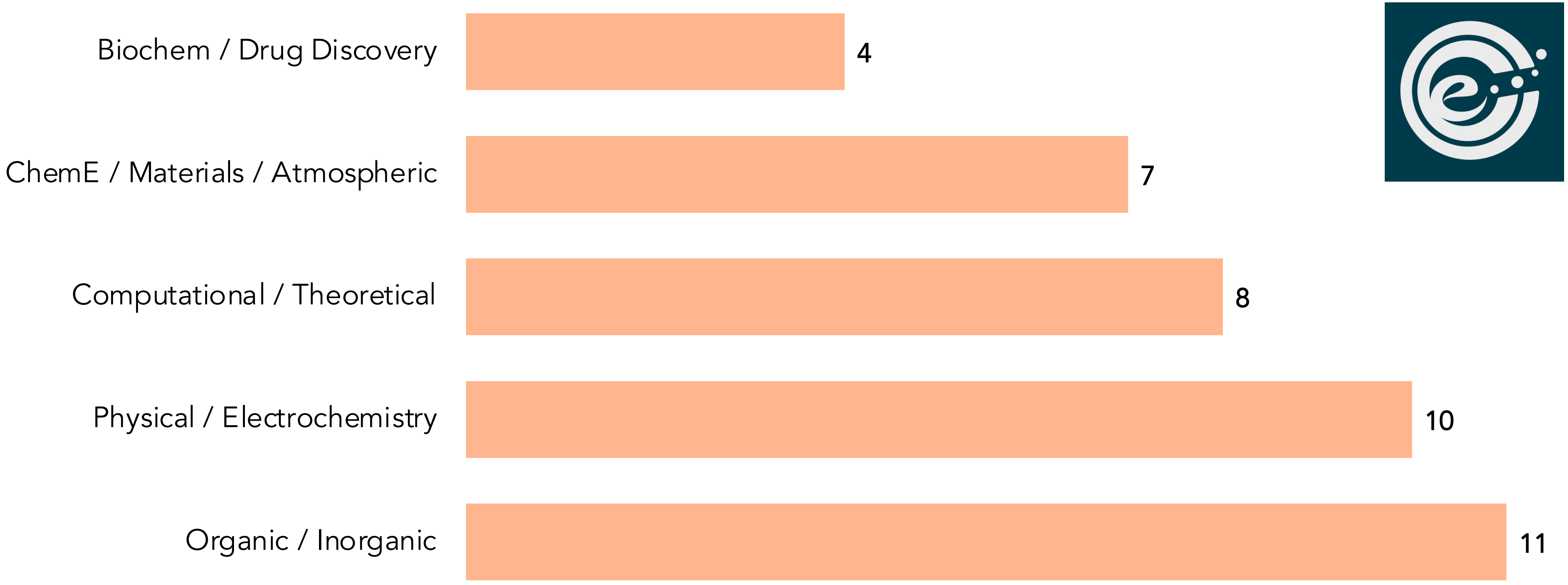
A. GURAJAPU
Hackathon Facilitation, MLIPs

H. JUNG
Tutorials Preparation

R. FOX
Logistics

T. KANO
Logistics

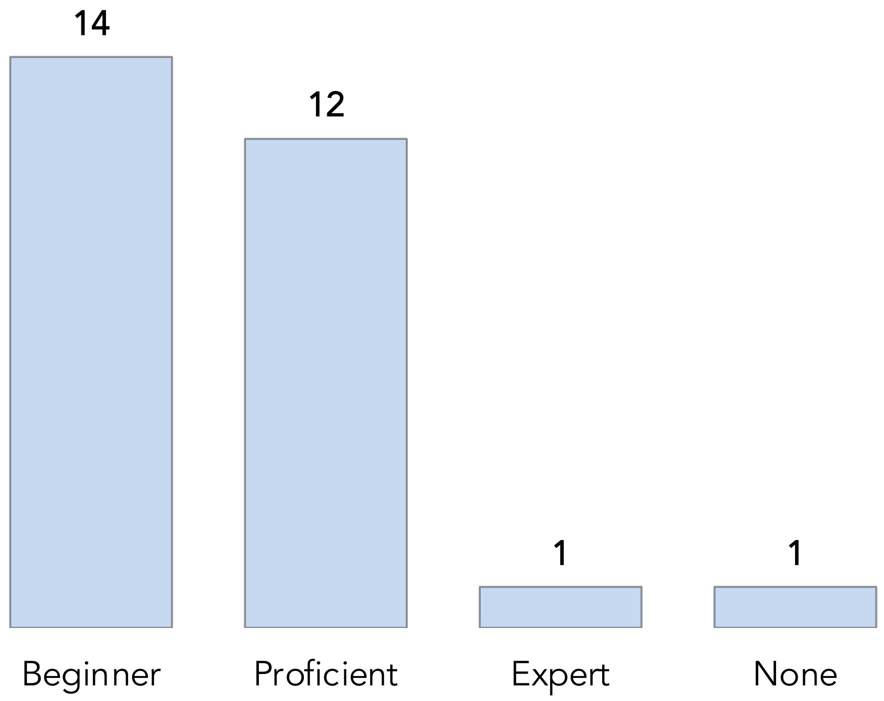
Attendance to the bootcamp



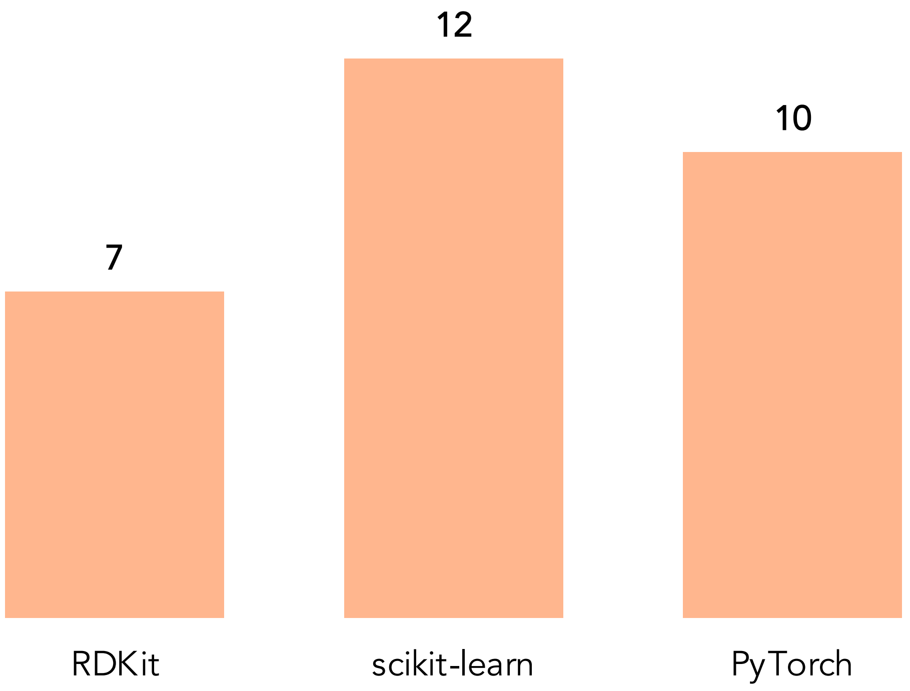
Classification done by Claude based on your responses to the Questionnaire

Coding Experience

Python experience

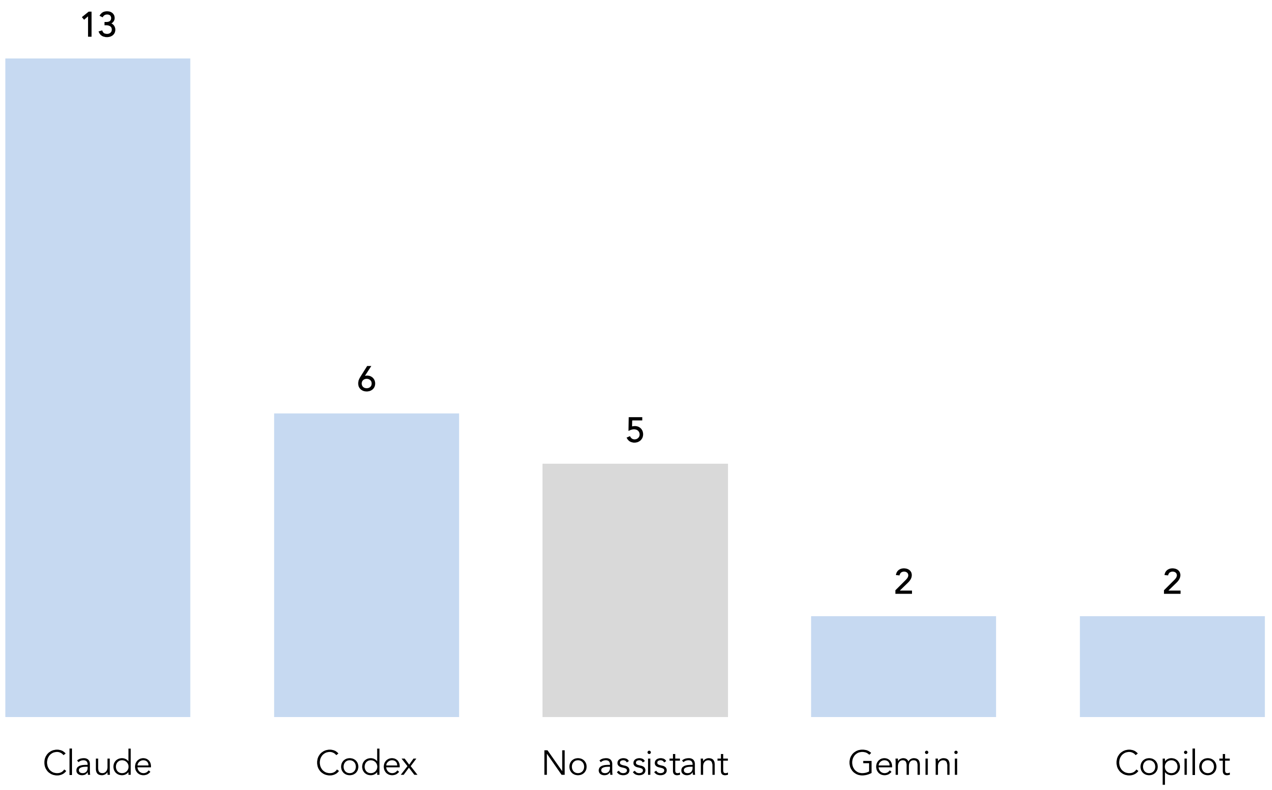


Familiarity with chemistry/ML tools



Based on Bootcamp Questionnaire

LLM assistant usage



Based on responses to the Questionnaire

~80%

already use an AI coding assistant day to day.

We will discuss how to use them efficiently

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Your expectations

“Automate the literature review & summarization”

“Machine-learned interatomic potentials (MLIPs) for MD”

“Ligand design & reaction-yield optimization”

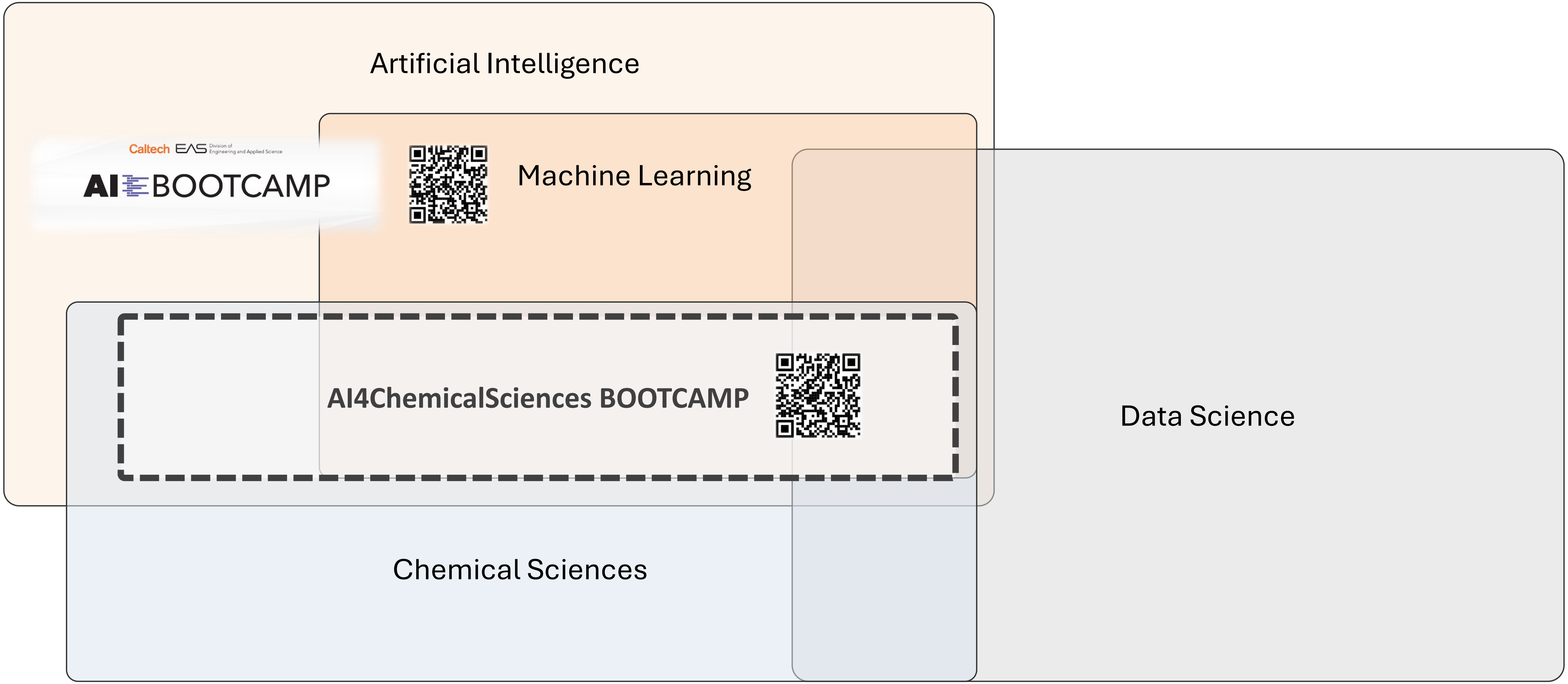
“Understand the “back-end” — which algorithm, why, how it's validated”

“Docking, virtual screening & binding-pose prediction”

“Multi-agent workflows / model selection guidance”

Based on Bootcamp Questionnaire

AI – ML – Data Science and Chemical Sciences



Building Machine Learning Models – What's needed?

DATA

The measurements and observations that ground everything a model learns.

REPRESENTATIONS

How a molecule or reaction becomes numbers a model can actually use.

ALGORITHMS

How we learn patterns from those numbers — and trust what comes out.

Models = understanding or black box for optimization

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AI Agents

DATA

The measurements and observations that ground everything a model learns.

REPRESENTATIONS

How a molecule or reaction becomes numbers a model can actually use.

ALGORITHMS

How we learn patterns from those numbers — and trust what comes out.

Agents can help accelerate the integration, they still need guidance

Scientific Data: Acquisition Bottleneck

- **Chemical data is expensive and slow to generate** especially for wet lab experiments
- **Datasets are small** compared to vision or language: hundreds to thousands of points, not billions.
- **Negative results and failed reactions are rarely recorded** — a strong, systematic reporting bias.
- Noise from instruments, batch effects, and inconsistent protocols across labs and instruments.
- Data is scattered across papers, ELNs, and proprietary silos — **rarely FAIR or ML-ready**.

Machine Readable Representations for Chemical Sciences

- **Molecules aren't naturally vectors** — there is no single canonical coordinate system for them.
- One molecule, many encodings: SMILES, InChI, SELFIES, fingerprints, 3D graphs — each keeps different information and hides different blind spots.
- Symmetries matter: permutation of atoms, rotation/translation in 3D, stereochemistry, tautomers.
- A representation that throws away the wrong information caps your model before training even starts.

Algorithms: choosing the model

- Which model fits your data: a linear model, a random forest, a graph neural net, a transformer?
- Which model fits your task: extrapolation, interpretability…?
- What model to use to rely on predicted uncertainty?

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AI Agents

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AI4Chemical Sciences: Learning Objectives

AI Agents

DATA

Dataset bias / distribution shift between benchmark and your real chemistry?

REPRESENTATIONS

What is the information in may representation?

Domain of applicability?

ALGORITHMS

Which algorithms to use?

Can we interpret the model?

Is model uncertainty reliable?

Bootcamp Roadmap

WEEK 1

Aug 10 – 14

Foundations of ML & Representations

Supervised learning, neural networks, molecular representations, GNNs / MLIPs, LLM assistants, generative models.

WEEK 2

Aug 17 – 21

Dataset Design & LLMs

Interpretability, design of experiments, Bayesian optimization, active learning, data stewardship, transformers, agentic AI.

Week 1 — Foundations of ML & Representations

MON	TUE	WED	THU	FRI
9:00 Intro & Logistics	9:00 Supervised Learning	9:00 MLIPs — GNNs	9:00 LLMs as Chemistry Assistants — RAG	9:00 Hackathon
13:30 Evolution of Molecular Representations	13:30 Neural Networks & Deep Learning	13:30 Learned Representations	13:30 Generative Models	

Week 2 — Dataset Design & LLMs

MON	TUE	WED	THU	FRI
9:00 Model Interpretability	9:00 Design of Experiments	9:00 Active Learning	9:00 Agentic AI for Chemistry	9:00 Hackathon
13:30 Data Stewardship	13:30 Bayesian Optimization	13:30 Transformers — text-based representations, tokenization	13:30 Hackathon	13:30 AI Readiness Wrap-up & Perspectives

Bootcamp Objectives

Goal for the two weeks:

- Basic literacy across data, representations, and algorithms
- Basics of when to start/not start an AI/ML project.
- Basics of how to get started coding wise.
- Build a network of chemists interested in AI4Science.

Tutorials objectives

- **Reproducible pipelines**: fixed seeds, versioned data splits, logged hyperparameters.
- Define **baselines.**
- **Code architecture that separates data, features, model, and evaluation**: test components independently.
- Readable code today is debuggable code tomorrow — including code an agent wrote for you.

Lecture are paired with Tutorial

9:00	→	10:45	→	13:30	→	15:15
Lecture	→	Tutorial	→	Lecture	→	Tutorial

Coffee 10:30–10:45 · Lunch 12:15–13:30 · Coffee 15:00–15:15

Fridays are full Hackathon days — apply the week's concepts to a real chemistry challenge with a presentation at the end.

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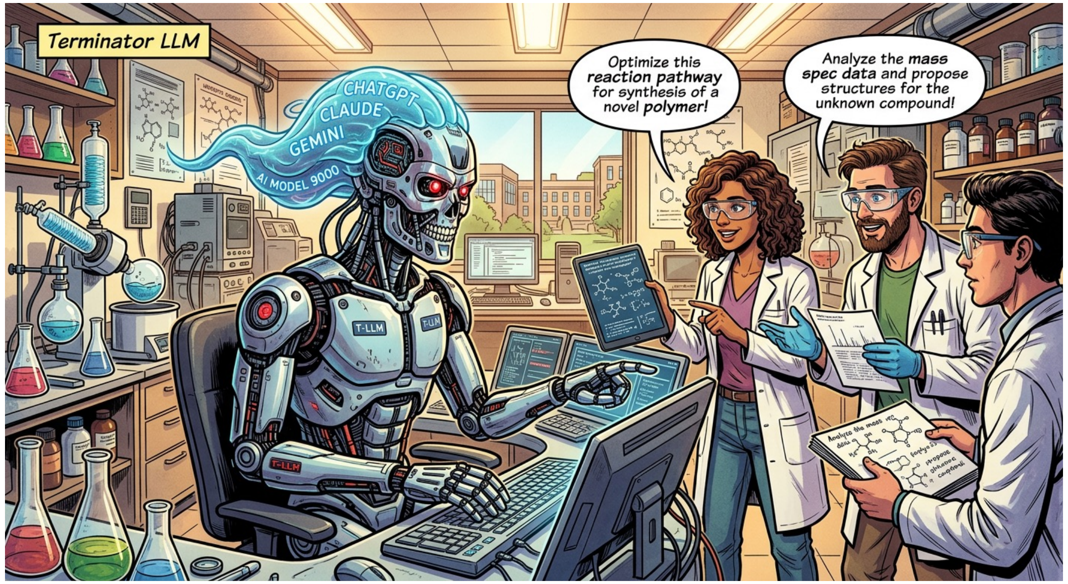

Coding agents are getting stronger, fast

↑ Capability

Agents now scaffold pipelines, debug, and write whole tutorials' worth of code in minutes.

↑ Cost

That capability is not free — token and compute cost scale with how much you let an agent run unsupervised.

↑ Need for direction

More power means you need to specify intent precisely and monitor what actually got done.

~80% of you already reach for Claude, Codex, Gemini or Copilot.

How to direct them well? How to keep track of what they produce?

Going faster without losing control

- Thursday, Week 2 — Agentic AI for Chemistry: ReAct, Coscientist, multi-agent workflows, self-driving labs.
- Build a Chemistry ReAct agent with PubChem lookup, RDKit tools, and a BO surrogate, with guardrails.
- Every hackathon: use agents deliberately — know what “good” output looks like before you delegate to one.
- Ground rule for the two weeks: review the diff, don't just trust the green checkmark.

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Where to find everything

Schedule

The live day-by-day plan — this page. Subject to small changes as the bootcamp goes.

Lectures

Slides & notes for every lecture, linked directly from the schedule.

Tutorials

Every hands-on notebook — the same ones you'll open each session.

Resources

Curated courses, papers & tools by topic — from ML fundamentals to self-driving labs.

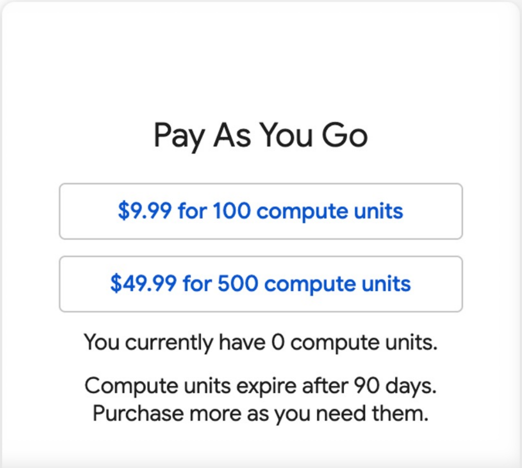
Claude and Colab Licenses

Claude Pro

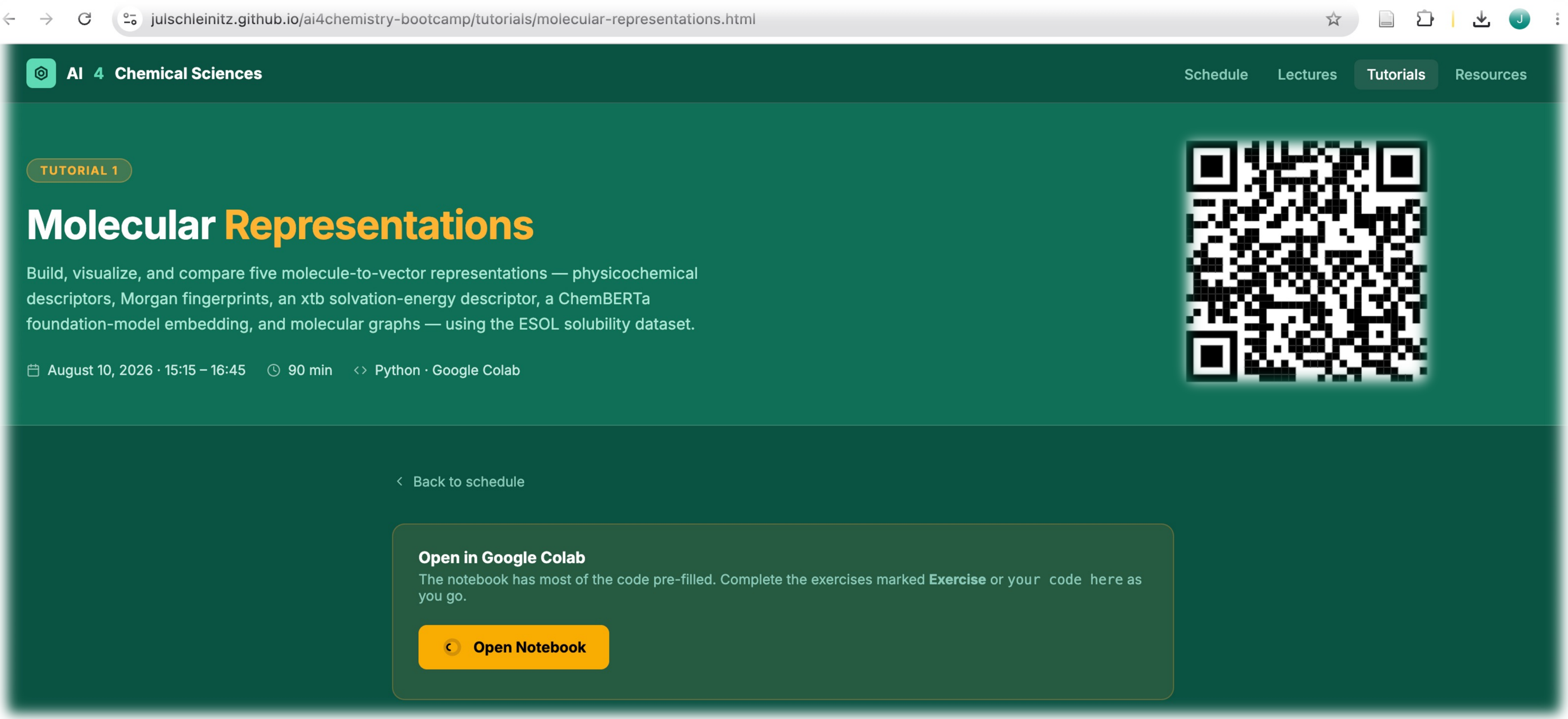
If you don’t have access, send me an email at jul@caltech.edu now

Colab Pro

If you don’t have access go to this web page using the QR code, select 100 compute units and raise your hand, Tomomi will help for the paiment



Checking Colab Interface



Let's meet each other

Next up, 10:45 — round-table introductions: your research background and what you're hoping to get out of these two weeks.

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